

Course: ENCN423/627 – Sustainable Energy Technologies

Module: Solar

Practice problems: Solar basics, array sizing calculations

Question 1.

Explain how the photoelectric effect creates mobile charge carriers within an n-p junction of two doped semiconductors

The photoelectric effect is the excitation of electrons to higher energy levels through the input of a photon. At an n-p junction of two doped semiconductors, this increase in energy level moves electrons from the e- saturated n-type donor to the e- depleted p-type acceptor. These mobile electrons are then transported via an electric field

Question 2.

What is the advantage of stacking multiple material layers (such as Perovskite and Silicon) within a single cell?

Different materials have different ranges of photon wavelengths which will provide the correct amount of energy to promote electrons, so material stacking can take advantage of a wider band of wavelengths within sunlight

Question 3.

If a cell's open-circuit voltage was tested in standard conditions to be 55 V, estimate the open-circuit voltage at a cell temperature of -5°C if $k_v = -0.15 \text{ V/}^\circ\text{C}$

$$V(T) = V_{ref} + (T_{cell} - T_{ref})(k_v)$$

$$V(-5) = 55 + (-5 - 25)(-0.15) = 59.5 \text{ V}$$

Question 4.

Why are inverters necessary for solar PV generation, and how do they achieve this function?

Inverters are necessary to convert the DC power generated by PV cells into AC power, the format of the wider electricity grid. This is done via pulse-width modulation to generate a wave which, when averaged, resembles a sine wave (this is then smoothed by other components such as inductors and capacitors)

Question 5.

List at least 5 potential loads the support structure of a freestanding PV module needs to be sized for during design

- Panels
- Live (inspection) loads
- Snow
- Wind (load and lift)
- Earthquake
- Mounting structure(s)
- Ice
- Etc.
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Question 6.

What purpose does a bypass diode serve within a PV array? In what situation(s) would one be useful?

A bypass diode provides an alternate current pathway to allow shaded (low current) components (panels or strings) to maximise possible generation in situations where one component/specific area is shaded or otherwise poorly generating (covered by debris etc.)

Question 7.

How can agrivoltaics systems increase average crop yields?

PV modules provide partial shading to crops below them, which can minimize the amount of time during which the solar irradiance is greater than the crop's photoinhibition point (point below which marginal increases in irradiance decrease photosynthetic volume)

Question 8.

If a house which receives 3.8 peak solar hours per day has 10 rooftop PV panels, each with a nominal power of 400 W and a performance ratio of 0.91, how much energy could they generate in a year?

$$P_{array} = N_{panel} \times P_{module} = 10 \times 0.4 = 4 \text{ kW}$$

$$E = P_{array} \times PSH \times PR = 4 \times 3.8 \times 0.91 = 13.8 \frac{\text{kWh}}{\text{day}} = 5037 \frac{\text{kWh}}{\text{year}}$$

Question 9.

A client is considering adding a floating PV array to their existing hydropower asset, present a brief discussion of the potential pros and cons of this endeavour

Floating photovoltaics can make use of space which is otherwise unutilized, but large installations can conflict with recreational uses of the reservoir and have large affects on water quality. Waves in the reservoir can be dangerous to modules if significant. Evaporation losses can be significantly reduced, and the cooling from the reservoir increases PV module efficiency. Large areas covered by PV panels can lead to constraints on minimum reservoir heights.